M/GWASC.1

KNX Gateway

Hardware Version: A

**Datasheet**

Issued: June 10, 2021

File Edition: B




Figure 1. KNX Gateway

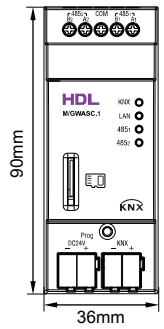



Figure 2. Dimensions - Front View

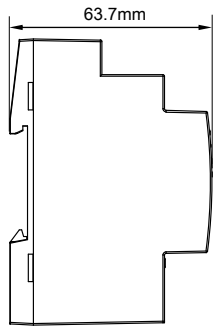



Figure 3. Dimensions - Side View

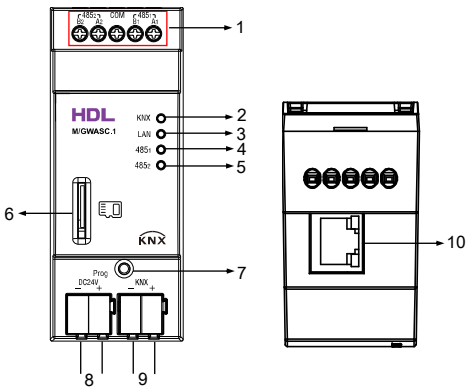



Figure 4. Components - Front View  Figure 5. Components - Side View

## Overview

KNX Gateway (see Figure 1) is a protocol conversion gateway based on KNX Bus communication, and enables data exchange with devices of different protocols.

Its main functions include:

- Device list: control feedback, update status
- Supports scene function: add scenes, *area capture scenes, *movie (dynamic) scenes
- Automation function
- Security function
- Clock: local clock, network clock, automatic synchronization of network time
- System detection, device status reporting: device online status, offline status, device abnormal information
- Drive management of third-party products
- Factory reset function
- Data backup and data recovery function

Protocol and software support:

- Internal communication protocols include: https, tcp, udp, etc., and standard mqtt access to third-party servers also supported
- Support intranet communication, remote access
- Support backup upload and download of APP, Cloud, *SD card, debugging software
- Auxiliary support: KNX Assistance Software, KNX ETS, ON+, Cloud server
- Online upgrade: upgrade gateway online via KNX Assistance Software, or upgrade other Bus devices through this gateway

Note: The functions with * are not yet open for use.

## Components and Operation

**Dimensions - See Figure 2 - 3**
**Components - See Figure 4 - 5**

1. RS485 connector
2. KNX indicator
3. Network status indicator
4. $RS485_1$ indicator
5. $RS485_2$ indicator
6. *SD card slot: supports up to 32G micro SD card
7. Programming button/indicator: 1) Short press the programming button, the indicator turns red for programming mode. 2) Keep pressing the button for 15 seconds to restore the gateway to factory settings.
8. Auxiliary power supply
9. KNX connector
10. RJ45 port

Note: The functions with * is not yet open for use.

## Note(s)

- Installation - Distribution box
- Device management - ETS can program and manage KNX TP1 standard devices through this device.
- KNX Bus voltage - 21~30V DC, no AC power supply allowed.

## Safety Precautions

- The installation and testing for the product must be carried out by HDL Automation Co., Ltd. or its appointed service agencies. The electric construction shall comply with local laws and safety regulations.
- The device should be installed with DIN rail in DB box. HDL will not be responsible for any consequence caused by the inexpert or faulty installation and wiring methods, which are not in accordance with the instructions contained in this operating instruction.
- Please do not privately disassemble or replace any parts of the product. Otherwise, it may cause mechanical fault, electric shock, fire or personal injuries.
- Please contact our after-sales departments or our designated service agencies for your maintenance service. Product failures caused by private disassembly are not subject to this warranty.

## Package Contents

KNX Gateway*1 / Datasheet*1



Figure 6

1

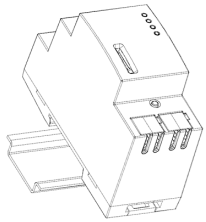


Figure 7

2

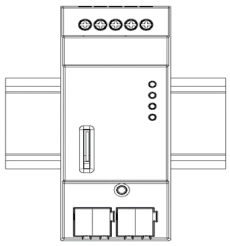


Figure 8

Figure 6 - 8. Installation

## Technical Data

| Basic Parameters | |
|---|---|
| Working voltage | 21~30V DC |
| Working current | 20mA/30V DC |
| Auxiliary voltage | 20~30V DC |
| Auxiliary current | 200mA/24V DC |
| Communication | KNX, RS485, Ethernet |
| Interface | KNX connector*1<br>Auxiliary power supply*1<br>RS485 connector*2<br>RJ45 port*1<br>SD card slot*1 |
| Hardware information | CPU H3@1GHz<br>4G RAM<br>8G Flash |
| Cable diameter of KNX terminal | 0.6 - 0.8mm |
| Operation system | Linux 3.2.0 Kernel |
| **External Environment** | |
| Working temperature | -5℃~45℃ |
| Working relative humidity | ≤90% |
| Storage temperature | -20℃~60℃ |
| Storage relative humidity | ≤93% |
| **Specifications** | |
| Dimensions | 90×36×63.7 (mm) |
| Net weight | 87g |
| Housing material | PA66 |
| Installation | 35mm DIN rail installation<br>(See Figure 6 - 8) |
| Protection rating (Compliant with EN 60529) | IP20 |
| **Approved** | |
| CE, RoHS | |
| KNX | |

## KNX Cable Guide

| KNX | KNX cable |
|---|---|
| - | Black |
| + | Red |

## Installation

**Installation - See Figure 6 - 8**

Step 1. Fix the DIN rail with screws.
Step 2. Buckle the bottom cap of the gateway on the edge of the DIN rail.
Step 3. Press the device on the DIN rail, slide it and fix it up until an appropriate position is adjusted.

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